COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structures and Analysis	**SECTION:**	CPE21S4
DATE PERFORMED:	July 22, 2026	**DATE SUBMITTED:**	July 22, 2026
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Seatwork 3.1 SINGLY LINKED LIST

1. PROCEDURE OUTPUT

SCREENSHOT

```
#include <iostream>
#include "singly_ll.h"

int main(){

    // Create Nodes: "S O P H I A"
    SinglyList <char>* head = nullptr;
    SinglyList <char>* second = nullptr;
    SinglyList <char>* third = nullptr;
    SinglyList <char>* fourth = nullptr;
    SinglyList <char>* fifth = nullptr;
    SinglyList <char>* sixth = nullptr;

    // Allocated Memory to the Head
    head = new SinglyList<char>;
    second = new SinglyList<char>;
    third = new SinglyList<char>;
    fourth = new SinglyList<char>;
    fifth = new SinglyList<char>;
    sixth = new SinglyList<char>;

    // Added the data to the mode
    head -> data = 'S';
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Testing the List Traversal Function:
S -> O -> P -> H -> I -> A
Testing the Insert at Head Function:
Q -> S -> O -> P -> H -> I -> A
Testing the Insert at any location Function:
Q -> S -> O -> X -> P -> H -> I -> A
Testing the delete a nodeFunction:
Q -> S -> O -> P -> H -> I -> A
Q -> O -> P -> H -> I -> A
Q -> P -> H -> I -> A
```

OBSERVATION ANALYSIS

- Linked list, as discussed, is a data structure that is made up of nodes, where each nodes stores a data and a pointer. Unlike the arrays that we did last time, this type of C++ function is used when the programmers insert and/or delete operations without a hassle. In this procedure activity, we used **ListTraversal()** to give us the access to visit every node inside the program. It starts with **head** and ends when it reaches **nullptr.** We also used **sllInsertEnd** to add a new node once the code reaches the last node. Next, we also used **sllDeleteNode**–to, of course as the name suggest, delete a

	node. - So I think after learning and digesting this information, Linked List can have advantages when it comes to adding and removing nodes without altering the whole code. It also targets one node without affecting the other, while, still, performing the code in the runtime.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Conclusion:

- To conclude this activity, learning these fundamentals can help me understand more the implementation of the fundamental operations and how to manipulate and manage the nodes to connect it correctly without altering the whole code and affecting the current progress.


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